

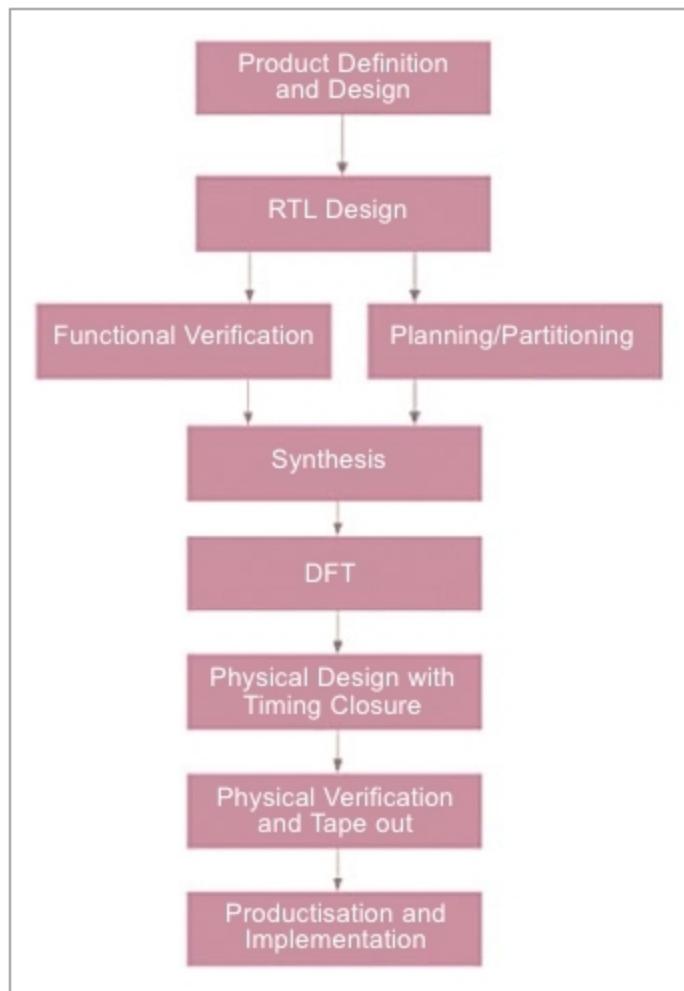


Fig. 3: Design flow of SSDs (Credit: www.cyient.com)

and driven the interface down a path of steadily increasing performance. Non-volatile memory includes block-accessible SSDs in card or drive form factors and byte-accessible devices such as NVDIMM. It operates out of RAM and utilises flash for backup to provide high speed. It is a non-volatile storage that plugs directly into the system memory bus.

Applications of SSDs

Jain says, "At Cyient we provide solutions based on customer requirement. As rail and transportation are being taken into perspective, SSDs are being used in train onboard memories, event and voice recorders, crash-protected memories, main processing units and drivers' display units, among others.

"Consider a driver's display unit that is used in the train cabin area for operator/train driver interaction/control and monitoring of train electronics. This display unit has been designed with

state-of-the-art microcontrollers suitable for this application. While selecting the microcontroller for such applications, we specially consider memory for cost and size estimation, and go for low cost, high endurance and high temperature grade and trade off among these metrics."

Cyient designs products that fulfill specific needs and requirements of specific domains and market segments like railway transportation. For the semiconductor industry, Cyient provides services and turn-key solutions for ASICs, SoCs, FPGAs design services, design verification and validation, physical IC design services, custom IC designs, custom memory development using static RAM and ROM. SSDs are suitable for high-load projects with numerous reading and writing operations.

SSS flash media like flash memory and micro flash memory cards are included in smartphones, MP3 players, laptops and desktops, personal computers, digital cameras, printers, tablets and servers.

All-flash drives have ports for flash memory cards. These transfer information from machine to machine.

Non-volatile flash memory cards

can be repeatedly erased, reprogrammed and rewritten without being removed.

SSS USB flash drives are used to move data and files from one computer to another. These connect to computers by way of USB ports. Flash drives are plug-and-play devices. There is no need to download any software or drivers before using these.

Advantages of SSDs over HDDs

SSDs do not have any rotating elements as these use memory chips. These make less noise, consume less power and are smaller than HDDs. SSDs are flash based and offer longevity, robustness, high temperature operating range, shock and vibration tolerance, and lower power consumption.

SSDs are also being used in data centres and for cloud storage. This is due to their superior performance, enhanced reliability, high endurance and low prices. Fast write speeds, low energy consumption, and higher program and erase cycles have made SLC SSDs the best option for enterprise use.

SSDs provide a simple user interface to all data centre employees so that they can easily interface with the SSAs. These provide a dedicated platform for administrators to use with ease, irrespective of the scale and size of their venture (small, medium, large or global).

IT staff and developers can easily execute operations such as replication, cloning and taking snapshots because of virtual machine level management. The ability to interface with a universal management platform enables control of all other aspects of the enterprise cloud, other storage types and data centres. SSDs can handle virtual server infrastructures, virtual desktop infrastructures and private cloud deployments. **EFY**

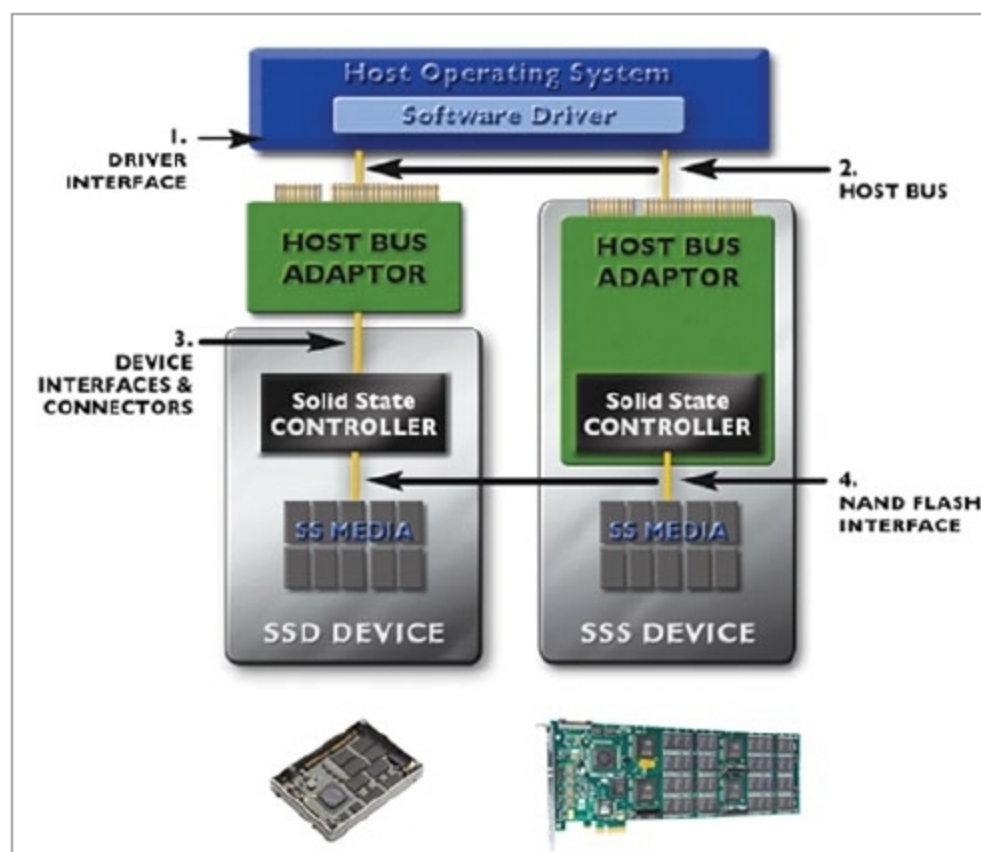


Fig. 4: Standards involved in the design and testing of SSS devices (Credit: www.snia.org)